

Mid Semester Examination September 2025

Name of the course: B. Tech (ECE)
Name of the Paper: Control Systems
Time: 90 Minutes

Semester: V
Paper Code: TEC 541
Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of sub questions.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	Determine the overall transfer function (using block diagram reduction technique) relating to C and R for the system whose block diagram is shown in Fig. below	CO1
(b)	Find the Laplace Transform of the differential Equations given below and hence evaluate the time solution of the same, (i) $2dx/dt + 8x = 10$ for $x(0+) = 2$ (ii) $d^2x/dt^2 + 2dx/dt + 2x = 0$ for $x(0+) = 0$ and $x'(0+) = 1$	
Q2	(10 marks)	
(a)	The transfer function of a system is given by $G(s) = K(s+8)/s(s+4)(s+5)(s^2 + 13s + 42)$ Determine the poles and zeros of the Transfer function. Plot the pole-zero configuration in s-plane.	
(b)	Develop signal flow graph for the block diagram shown in fig. below and determine C/R using Mason's gain formula.	CO2
Q3	(10 marks)	
(a)	A closed loop transfer function of a unity feedback control system is given by $C(s)/R(s) = (Ks+\beta)/(s^2 + \alpha s + \beta)$. Determine the steady state error for unit ramp input.	CO3
(b)	Derive expression of time response of first order system subjected to (i) unit step input (ii) unit ramp input. Plot their transient and steady state.	
Q4	(10 marks)	
(a)	A unity feedback control system has an open loop transfer function $20/(s+1)(s+2)$ Find rise time, percentage of maximum overshoot, peak time and setting time for 2% and 5%.	CO4

(b)	Explain various types of control systems and find their expression for steady state error.	
Q5	(10 marks)	
(a)	Explain Armature Control D. C. Servo Motor and derive their expression for Transfer function with respect to angular displacement and Angular Velocity.	CO2/CO3
(b)	Draw the signal flow graph of the system equation given below and find X_3/u_1 using Mason's gain formula. $X_1 = a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + b_1u_1$ $X_2 = a_{21}X_1 + a_{22}X_2 + a_{23}X_3 + b_2u_2$ $X_3 = a_{31}X_1 + a_{32}X_2 + a_{33}X_3$	